

BIOLOGY

Notable Faculty & Student Achievements

2025 Faculty Awards: Dr. Kim McArthur was awarded the Junior Faculty Teaching Award, and Dr. Maria Todd was awarded the Faculty Advising Award.

February 2026

Professor of Psychology **Fay Guarraci**, along with several students and alumni, published an article titled "Chronic testosterone enanthate or estradiol benzoate affects sexual behavior and motivation in Long-Evans rats tested for partner preference" in *Physiology and Behavior*. This work was supported by summer research programs (SCOPE '24 and SURF '25) and conducted by psychology and biology capstone students. **Raylee Bowling** '27 is first author, followed by **Chance Bauer** '26, **Iva Irabor-Ighedosa** '26, **Kai Wilson** '26, and **Anna Pavlova** '28. Alumna **Alex Avila** '25 and visiting scholar Kole Petersen, a student from Colorado College, were also co-authors. This project tested a model of gender-affirming hormone therapy in an animal model.

Associate Professor of Psychology **Carin Perilloux**, along with psychology graduate **Madeline Bollinger** '24, psychology graduate and Coordinator of the Professor John Score Learning Commons **Laurel Mulkey** '24, and biology graduate **Vy Nguyen** '23, have published a paper titled "Hooked on a feeling: Psychological and physiological responses to ASMR triggers" in the journal *Technology, Mind, and Behavior*. This article is based on three years worth of SCOPE projects and is published with open access here.

Assistant Professor of Biology **Sunny Scobell** gave an invited seminar to the Department of Biology at Texas State University, titled "The physiological regulation and evolution of male pregnancy," on November 14.

Assistant Professor of Biology **Sunny Scobell** gave an invited seminar to the Biology Department at Trinity University, titled "The mysterious male brood pouch: Structure and function of male pregnancy in seahorses and pipefish," on November 10.

Professor of Anthropology **Melissa Johnson** and her Summer 2025 SURF students

students presented their papers: anthropology and feminist studies double major **Dina Gaxiola** '28 presented "The Case of Jane Trapp: Race, Gender, Slavery and Freedom in Early British Central America," psychology and sociology double major **Sophia Hernandez** '28 presented "From Enslavement to Influence; Gendered Agency, Racial Boundaries, and the Story of Ariadne Broaster," Dr. Johnson presented "Living across lines of freedom in the early history of Belize: The origins of the Bonner family," and biology and environmental studies double major **Shae Whitney** '28 presented "Maria Middleton – Slave, Criminal, Transported Convict: Understanding Gender, Race and Freedom across the British Colonial World." The students may have been the only undergraduates presenting papers at this conference. The papers were well received, with one audience member sharing that this was the most interesting panel they had attended.

October 2025

Supported by funds from the Garey Chair in Biology and the Mellon Grant for Publicly Engaged Scholarship, Professor of Biology **Romi Burks** gave a talk and tasting at the Northwest Chocolate Festival, titled "Get Origin(al): Educated by chocolate in Brazil and Peru." Dr. Burks shared her professional development experiences abroad this summer learning about supply chains and post-harvest practices in two important origin countries that grow *Theobroma cacao*, the tree that provides the beans from which we make chocolate. She also participated in the "Unconference," a networking opportunity for professionals in the chocolate industry.

June 2025

Alumna **Esmeralda Rosas** '24 recently published her biology capstone as part of an applied journal section called "Management in Practice" within the journal *Management of Biological Invasions*. Co-authored with fellow apple snail lab alumna, **Cynthia Bashara** '23, this study reports a four-year monitoring effort with Austin community member David Christie that resulted in a local eradication of invasive snails. Success stories in invasion biology often get overlooked, but this paper, titled "Winning the shell game: environmental DNA (eDNA) confirms local control of the invasive apple snail, *Pomacea maculata*" provides a template for others to follow. A downloadable PDF of the paper is available [here](#). Made possible by devoting time during a recent sabbatical, this publication represents the second outcome of the ongoing collaboration between Professor and Garey Chair in Biology **Romi Burks**, and **Matthew Barnes** '06, Associate Professor at Texas Tech University.

May 2025

Kinesiology alumni **Corban Ruiz**'s '23 "fun" COVID student research project turned independent study with Associate Professor of Kinesiology **Ed Merritt**, which morphed into a research assistantship data management role for biology major and kinesiology minor **Iliana Hernandez** '27, was published in the *International Journal of Exercise Science*. The paper, "Age of Anaerobic, Aerobic, and Skill-Based Olympic Athletes 1988 – 2024" provides insight into the development of and, potentially, the onset of age-related declines in human metabolic systems. The good news for most of us is that there has been at least one 65-year old who has competed in the Olympics since 1988 in a non-equestrian event! The paper can be read [here](#).

Biology major and kinesiology minor **Iliana Hernandez** '27 presented research findings from the recently completed clinical trial "Exercise Training Adaptations in Middle-Aged Adults With and Without Post-Exercise Peanut Consumption" at the American Physiology Summit in Baltimore, MD on April 26. Co-authors on the study are kinesiology major **Erica Otto** '25, Southwestern alumni **Matthew Bierwirth** '24, **Rachel Chiella** '24, **Barrett Knapp** '24, and **Zachary Moon** '24, and Associate Professor of Kinesiology **Ed Merritt**.

March 2025

The weekend of February 28 – March 2 brought scientists from all over Texas to McLennan Community College and Baylor University in Waco, for the 145th Texas Academy of Sciences meeting. Ten students represented the Biology Department, including **Gage Mallo** '26, **Johun Reyes** '26, **Annika Tracy** '25, **Murphy Jacobie** '27, **Madeleine Thomas** '26, **Kylie Allemeier** '26, **Katelin Pilarski** '26, **Sarah Berver** '26, **Alanna Guerrero** '25 and **Jordan Emerson** '27.

Based on her collaboration with Assistant Professor of Biology **Kim McArthur**, Annika gave a talk, titled "Location-independent axon pathfinding in the hindbrain of larval zebrafish," that earned the first place undergraduate talk award for the neuroscience section.

Madeleine presented a poster with her research mentor, Assistant Professor of Biology **Sunny Scobell**, titled "A morphological analysis of the dopaminergic pathway in the brain of a male pregnant pipefish." Dr. Scobell also mentored Jennifer Smalz, a student from her previous institution (UD: The University of Dallas), who gave a talk titled "Investigating male pregnancy in Gulf Pipefish using scanning electron microscopy to describe the anatomy of the brood pouch over the reproductive cycle." Ms. Smalz' talk earned the first place undergraduate award in the systematics and evolutionary biology section. Within that

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prolactin during reproduction in a fish with male pregnancy."

Murphie presented a microbiology focused poster, titled "An affinity analysis of *Aliivibrio fischeri* and *zooxanthellae*" with the mentorship of Director of First Year Biology Laboratories **Stacie Brown**.

Working with Chair and Garey Professor of Biology **Romi Burks**, Gage presented a poster, titled "You better Belize they're different: Phylogenetic analysis and species identification of native apple snails in Belize," that garnered a first place best poster award in the systematics and evolutionary biology section. His poster co-author Johun secured first place in the Undergraduate Research Award Competition, complete with a \$2,000 prize to continue the work, with his proposal, "Beneath Belizean waters: Discovering diversity of native apple snails in northern and southern Belize."

Also from the Burks lab, Kylie, Sarah, and Katelin shared their recent troubleshooting efforts in the lab in a poster titled "Extraordinary extraction efforts: Experiments to enhance DNA extraction for tissues of apple snails of conservation interests" that also used collections from Belize that occurred as part of the 2024 study abroad experience.

Dr. Burks also gave an oral presentation in the science education section about the recent undergraduate research experience that she developed in BIO50-222, Methods in Ecology and Evolution. Her talk, titled "Methodological microplastics: Development of an undergraduate CURE to quantify abundance of microplastic fibers in a local stream," came out of conversations with microplastics expert Andre Felton from the University of Texas at San Antonio.

January 2025

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de aves en ecosistemas antropizados y no antropizados del corredor seco nicaraguense," in a special edition of *Revista Científica Estelí* in December. He also presented a poster titled "Golden-cheeked Warbler habitat preferences in Nicaraguan Highlands," in conjunction with co-authors representing Universidad Nacional Autónoma de Nicaragua, Université de Sherbrooke, and Virginia Tech, at the Texas Conservation Symposium in January.

November 2024

Assistant Professor of Biology **Kim McArthur** authored a spotlight article recently accepted in the journal *Trends in Neurosciences*. The article, titled "Cranial motor neuron input specificity refined by activity," provides a critical analysis of a recent primary research article in her area of expertise (Kaneko et al. 2024, "Position-independent functional refinement within the vagus motor topographic map"). The spotlight and research article both address the "big picture" question of how motor neurons in the vertebrate hindbrain receive appropriate synaptic inputs during early motor circuit development.

October 2024

Assistant Professor of Biology **Kim McArthur** co-authored a research article recently

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(*Danio rerio*)," uses a combination of single-cell labeling, in vivo fluorescence microscopic imaging, and 3D tracing to provide evidence that early synaptic input to motor neurons may be determined by the relative location of their dendrites in the developing hindbrain. This work contributes to our growing understanding of the importance of early spatial organization for the development of neural circuits.

September 2024

Buoyed by the momentum built on sabbatical, Garey Chair of Biology **Romi Burks** continued her work on the intersection between chocolate and liberal arts education by sharing her knowledge as the featured STEM kick-off speaker at Schreiner University on September 4 with a talk titled "Chocolate Covered Science: Connect what you love to eat with what you love to study." Following that event, Burks attended the Dallas Chocolate Festival on September 7, where she contributed to a panel discussion on the crisis in the chocolate industry, along with entrepreneur Kate Weiser of Kate Weiser Chocolates and Jimmy Steward of Guittard Chocolate. Afterwards, Burks gave a solo presentation titled "Educated by Chocolate: How to build your chocolate knowledge one bite at a time." Burks has also been invited to give this talk again at the Northwest Chocolate Festival, along with a second presentation titled "Separating Fact from Fiction in the Science of Chocolate." These lectures will take place the first weekend of October.

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paper, titled “Snail slime in real time: Challenges in predicting the relationship between environmental DNA and apple snail biomass,” describes how extrapolations can fail between snail abundance and the amount of genetic material produced by apple snails under cold conditions. The results have implications for estimating the size of non-native invasive populations. Student co-authors include **Cassidy Reynolds '24**, **Esmeralda Rosas '24**, **Cynthia Bashara '23**, and **Lillian Dolapchiev '23**, along with alumni collaborator Dr. **Matthew Barnes '06** from Texas Tech University and Dr. Chris Jerde from UC Santa Barbara, who provided assistance in data analysis. The open access paper can be read here, as #6 in the table of contents of the journal issue.

August 2024

Professor Emeritus of Biology **Ben Pierce** was appointed to the Bell and Coryell Counties Biological Advisory Team, which will assist in the development of a regional habitat conservation plan for these two Texas counties.

April 2024

On April 25, Visiting Assistant Professor of Biology **Kyle Wilhite** and his Ph.D. advisor, Mike

In 2023, Professor of Biology **Romi Burks** contributed to a United States Department of Agriculture - Agriculture Research Services (USDA-ARS) effort to communicate the latest science on the invasive apple snail that occurs across the southeastern region of the United States. The content of that Apple Snail Workshop, entitled "Apple Snails: Discussion of past problems and future solutions for an emerging pest in the United States agriculture/aquaculture," now appears published in *World Aquaculture*, the magazine for the World Aquaculture Society. Recent Burks' Lab researcher, now professional scientific illustrator, **Lauren Muskara '21** provided the photograph of an apple snail laying eggs on a pylon that the authors chose for the cover of the article. The full article can be read on page 46 here.

January 2024

Visiting Assistant Professor of Biology **Kyle Wilhite** published a paper titled "Ripple effects in a communication network: anti-eavesdropper defense elicits elaborated sexual signals in rival males" in the journal *Proceedings of the Royal Society B* on December 20, 2023. This research was a multi-national collaboration with Purdue University, The University of Texas at Austin, VU University in the Netherlands, the University of Tennessee Knoxville, the University of Antioquia in Colombia, and The Smithsonian Tropical Research Institute.

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Assistant Professor of Biology **Kim McArthur** presented a virtual poster at the annual meeting of the Society for Neuroscience in November 2023. Her poster, titled "Mapping the dendrite topography of facial motor neurons in larval zebrafish," presented preliminary results from her sabbatical research, gathering evidence to test the hypothesis that the relative positioning of a neuron's dendrites can determine which synaptic inputs that neuron receives— thereby determining its functional role in a neural circuit.

October 2023

Professor of Biology **Romi Burks** continues "chocolating" her way through her sabbatical. On October 6th, she gave an invited talk at the Northwest Chocolate Festival in Seattle, the largest professional gathering of fine chocolate professionals. Taken place on the Fine Chocolate Industry Association's stage, the talk "you say cacao, I say cocoa" chronicles her work on and vision for the Fine Chocolate Glossary project. Burks left for 3 weeks in London, where she will judge within the UK's premier organization, The Academy of Chocolate. All of this travel and networking will contribute to her book project "Journey into Chocolate," coauthored by Indi Chocolate owner and entrepreneur Erin Andrews.

Professor of Biology **Romi Burks** and chocolate pedagogy enthusiast gave two invited presentations this past weekend at the Dallas Chocolate Festival, which featured a theme of "The Dream of Chocolate." On Saturday, Burks spoke about her work as the Chair of the Fine Chocolate Glossary project within the Fine Chocolate Industry Association with a title - Chocolate "Definitions" - the dream of developing a common language. On Sunday, Burks stepped 48 guests through "How to Taste Chocolate like a Competition Judge" talk where participants learned about criteria for fine chocolate. When not giving presentations, Burks staffed a booth to talk to the attendees about the Fine Chocolate Glossary. Pictures can be found on her ProfRomi Instagram and other social media platforms.

August 2023

Assistant Professor of Biology **Jennie DeMarco** attended and presented her research on using compost and native seeding to restore ski slopes and sequester soil carbon at the 2023 Ski Conservation Summit held at Copper Mountain Ski Resort in Frisco, Colorado. Also in attendance were Southwestern University students Logan Antone '24, Cooper Phillips '24, Blaine Ten Wolde '25, Hailey Vickich '25, and Olivia Johnson '26. The summit and our research were covered in several media outlets: 5280 Denver's Mile High Magazine, CBS Colorado, Summit Daily, and SAM Magazine.

Professor of Biology **Max Taub** and Marcelo Salazar-Barragan '23 published the paper "The Effects of Elexacaftor, Tezacaftor, and Ivacaftor (ETI) on Blood Glucose in Patients With Cystic Fibrosis: A Systematic Review" in the Cureus Journal of Medical Science.

Professor of Biology **Romi Burks** attended the Fine Chocolate Industry Association meeting in New York City on Saturday, June 24th. She led a "Lunch and Learn" interactive presentation on the Fine Chocolate Glossary project during that time. Visual art major Ryan Tanner '25 designed the new logo for the glossary and helped craft two display posters as part of her summer internship. The FCIA session titled "Chocolate: I do not think it means what you think it means" plays off the famous quote about the inconceivable from The Princess Bride. Drawing on that, Burks wrote a blog post about her thoughts on taking over the leadership of the Glossary project. This open-access resource recruits professionals in the chocolate industry to author entries based on their experience and research, and each entry can then receive feedback and revision. The project seeks to establish a common language within the world of fine chocolate. Being part of the project and networking within the chocolate industry contributes directly to Burks' sabbatical plans to write a mainstream book on chocolate.

April 2023

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undergraduate research-based laboratory course that cultivates an inclusive and supportive learning community and helps students develop a sense of belonging through mentorship and peer-to-peer learning. The course is centered on a new framework they developed called crisscrossing laboratory experiences (CCLE), where first-year students were placed into two interdisciplinary research cohorts. Halfway through the course, the cohorts swapped research projects to establish reciprocal peer learning partnerships allowing students to be both learners and teachers.

February 2023

Professor of Biology Romi Burks engaged in a number of off-campus presentations in February about her research on apple snails and environmental DNA and also her passion and philosophy of teaching through chocolate. The chocolate events occurred in person, while the science talks occurred as part of a federal government-sponsored workshop and a Texas-based science podcast. On February 12: Not Just a Casual Love Affair: How to Cultivate a Real Relationship with Chocolate; Lecture and tasting at the Rockport Center for the Arts, February 17: Biology & Chemistry of Chocolate; Lecture and tasting for Westwood High School Enrichment Program, February 21st: Now You See Them, Now You Don't? Using eDNA to confirm removal of invasive snails by local agency; APHIS-USDA 2023 Apple Snail Workshop, and February 24th: Science Stories Podcast (Live on community radio station KZSM) with Dr. Mateo Garcia, a postdoctoral researcher at Texas State University.

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work was part of Dr. DeMarco's former graduate student's thesis and focuses on how we can manage pasture lands to sequester carbon into the soil as a climate change mitigation strategy. The paper can be accessed [here](#).

Assistant Professor of Biology **Kim McArthur**, with co-authors Emma Astad '21, Emmett Griffin-Baldwin '22, Bria Tovar '22, and Tori Tovar '22, wrote a research article recently accepted in the Journal of Comparative Neurology. The article, entitled "Early development of respiratory motor circuits in larval zebrafish (*Danio rerio*)," uses a combination of behavioral analysis and functional calcium imaging to reveal the developmental time course of early synaptic inputs from respiratory, central pattern-generating circuits to the cranial motor neurons that generate coordinated breathing behaviors. This work provides critical foundational information about the development of an emerging model circuit in developmental neurobiology, setting the stage for future work probing the mechanisms of synapse development.

November 2022

Professor of Biology and 2022 London Faculty **Romi Burks** gave an invited presentation on October 22 on chocolate education entitled "Making the word chocolate mean more to everyone" at the first edition of the Latin American International Festival of Chocolate and Cocoa in Europe. This event took place in partnership with The Chocolate Story Museum

Professors of Biology **Maria Cuevas** and **Maria Todd** published a paper titled "Microarray Analysis Reveals Overexpression of both Integral Membrane and Cytosolic Tight Junction Genes in Endometrial Cancer Cell Lines" in the Journal of Cancer. The study reports for the first time a comprehensive analysis of 84 tight junction network genes in a panel of endometrial cancer cell lines. The authors identified three genes that were consistently deregulated in endometrial cancer, thus providing potential candidates for the development of both diagnostic markers and novel therapeutic approaches for endometrial cancer.

October 2022

Assistant Professor of Biology **Jennie DeMarco** was an invited speaker at the September 23 Baylor University Department of Biology Seminar, where she presented her research titled "From Transect to Landscape: Evaluating the Role of Wet Meadow Restoration in Climate Change Mitigation and Resilience"

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Change Mitigation and Resilience.”

September 2022

Professor of Biology and Lillian Nelson Pratt Chair **Ben Pierce** published a paper titled “Relative Tail Width as an Indicator of Body Condition in Central Texas *Eurycea* Salamanders” in the August 2022 issue of *Herpetological Conservation and Biology*.

August 2022

A new research project by Assistant Professor of Biology **Jennie DeMarco** was featured in the newspaper *Summit Daily*, which serves the Summit County, Colorado, area. The article, titled “Copper Mountain Resort Begins 10-Year Carbon Sequestration Study,” highlighted the collaboration between Southwestern University, the Copper Mountain ski resort, and Peak Ecological Services, LLC, to investigate the impact of ski slope restoration as a nature climate solution.

Assistant Professor of Biology **Jennie DeMarco** and Southwestern students **Gabrielle Garza '22**, **Guadalupe Sanchez '23**, and **Christine Vanginault '23** attended a conservation summit held July 27 at the Copper Mountain ski resort in Colorado. DeMarco and her collaborators—Jeff Grasser, efficiency manager at Copper Mountain, and Rea Orthner, botanist/ecologist with Peak Ecological Services, LLC—presented their research project titled “Ski Slope Restoration as a Nature Climate Solution.” The goal of the summit was to enhance collaboration and provide an opportunity for resort operators, land managers, and researchers to share innovative methods for enhancing the conservation of local ecosystems impacted by recreation.

May 2022

Assistant Professor of Biology **Jennie DeMarco** was awarded \$14,580 from the Upper Gunnison River Water Conservation District for her grant proposal titled “From Transect to Landscape: Investigating the Role of Remote Sensing Tools to Monitor Soil Moisture with Wet Meadow Restoration in the Gunnison Valley.”

Assistant Professor of Biology **Jennie DeMarco** presented her research titled “Evaluating the Carbon Sequestration Potential and Drought Resilience with Wet Meadow Restoration under a Changing Climate” at the High Altitude Revegetation Committee and Society for Ecological Restoration–Rocky Mountains Chapter 2022 Conference held April 13–14, in Fort

April 2022

Assistant Professor of Biology **Kim McArthur** gave an invited virtual presentation on equity in grading to the Union College (Schenectady, New York) faculty on March 22. The presentation, titled "Rethinking Traditional Grading, Working Towards Equity," provided a framework for considering the impact of traditional grading practices on student equity, with suggestions for reconsidering and fully aligning assessment practices with course learning objectives.

March 2022

During the last weekend of February, Professor of Biology **Romi Burks** and Professor of Biology **Ben Pierce** took eight biologists to the Texas Academy of Sciences (TAS) Annual Meeting in Clear Lake, Texas. Southwestern student contributions included two posters and two oral presentations as well as an exciting second-place team finish in "Science Jeopardy."

- Two of Pierce's research students, **Sydney Cole** '23 and **Claire Bason** '23, won second place for best undergraduate poster in the terrestrial ecology and

coauthors included **Emma Resterson** '25 and **Gina Ramenz** '25.

- In the freshwater science section, **Lillian Dolapchiev** '23 gave a talk titled "Filter Me ... If You Can: Using Size Fractionation to Separate, Measure, and Determine the Size of *Pomacea maculata* eDNA." Her coauthors included **Cynthia Bashara** '23, **Matthew Barnes** '06, and Burks. Dolapchiev earned first place for best undergraduate oral presentation within the freshwater science section.
- In the same section, Bashara gave an oral presentation titled "Snail (*Pomacea maculata*) Days of Summer: Associations Between Reproductive Output, Snail Removal Efforts, and Environmental DNA (eDNA) Concentration," which included Dolapchiev, Barnes, Burks, and Chris Vaughn from the San Antonio River Authority as coauthors. Bashara took the second place award in the category.
- Together, Bashara and Dolapchiev presented their specific research objectives completed over the summer during SCOPE as a poster presentation titled "Stop Escargo in San Antonio: Developing Best Methodology for Detecting *Pomacea maculata* Using Environmental DNA (eDNA)." This poster won second place for best undergraduate poster in the freshwater science section.
- Two more research students of Burks, **Kate Henderson** '25 and **Abby White** '25, also presented a poster in the freshwater science section titled "Keep Austin Snail-Free: Ongoing Removal of *Pomacea maculata* and Evaluation by eDNA." The poster was coauthored by Bashara, Dolapchiev, and Dave Christie, who owns a home in Austin that has been invaded by apple snails. Henderson and White put together this poster based on just a semester of lab involvement.
- In addition to collaborating with Burks and coauthoring presentations with Bashara and Dolapchiev, Barnes, an associate professor at Texas Tech University, served as vice president of the academy and transitioned into his 2022–2023 role as president-elect. He will oversee the program at next year's TAS meeting at San Angelo State University. His own undergraduate and graduate students from Texas

Feburary 2022

Professor of Biology **Romi Burks** recently participated in a podcast called *Conversations About Cocoa* where she discussed her educational journey to becoming a "chocolate expert." Host Lauren Heineck, a chocolate educator and moderator of the Facebook group for chocolate professionals Well Tempered, spoke with Burks about her mission in education, the chocolate industry, and how studying apple snails translates into understanding more about the genetics of cacao (the plant from which chocolate comes) and vice versa. You can access the podcast on Heineck's website.

January 2022

Five Southwestern students had the opportunity to extend their coursework and research experience beyond the classroom with poster presentations at the Texas Conservation Symposium, which was cosponsored by Southwestern and the Williamson County Conservation Foundation. The students all had the opportunity to interact with keynote speaker Kelly Ramirez, assistant professor at the University of Texas at El Paso and cofounder of 500 Women Scientists. Three of the presentations built on work the students

- **Katherine Montgomery** '23: "The Blackland Prairies in 2050: Never Lost, Just Too Often Forgotten"
- **Lauren Wheat** '23: "Edwards Plateau 2050: Need for Increased Conservation of Golden-cheeked Warbler and Black-capped Vireo Nesting Habitat"
- **Nicole Ratjak** '22: "2050 at the Beach? Conservation Concerns for the Future of the Texas Gulf Coast and Prairies Ecosystem"

In addition, two students mentored by Burks in the Molecular Aquatic Ecology Lab, **Lillian Dolapchiev** '23 and **Cynthia Bashara** '23, presented their research from the 2021 SCOPE program titled "Escar-go to San Antonio: Using Environmental DNA to Detect the Non-native Invasive Species *Pomacea maculata*."

Both Montgomery and Dolapchiev received recognition for outstanding presentations. **Michael Gervasi** '23 also had his poster, "Trans Pecos 2050," on display. You can view the ecoregions posters on Burks's website.

Professor of Biology **Ben Pierce** and Assistant Professor of Biology **Jennie DeMarco** also gave presentations on their research, "Relative Tail Width as an Indication of Body Condition in Central Texas *Eurycea* salamanders" and "Invasive Species Litter Quality Alters Ecosystem Function through Enhanced Litter Decomposition Independent of Drought Conditions," respectively. Pierce works each year to organize this symposium on behalf of Southwestern.

July 2021

Assistant Professor of Biology **Kim McArthur** presented her research at the annual meeting of the Society for Developmental Biology on July 14, 2021. In her virtual poster, titled "Behavioral Analysis of Respiratory Circuit Development in Larval Zebrafish," McArthur

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stages in neural circuit development as zebrafish develop outside of their mothers where brain cells can be observed under the microscope using noninvasive methods.

April 2021

Business and biology major **Andrew Vergote** '21 gave a talk titled "Novel Bioinks: The Gateway to Bioprinting Complex Biological Tissues" at the 2021 BBB South Central Virtual Regional Convention. The talk resulted from research that Vergote completed with Assistant Professor of Physics **Cody Crosby** in spring 2021. They plan to continue this research as a funded Faculty–Student Project this summer.

August 2020

During spring and summer 2020, Professor of Biology **Romi Burks** was selected as a member of a Faculty Mentoring Network (Make TRUBLE) within the Quantitative Undergraduate Biology Education and Synthesis (QUBES) consortium. QUBES represents a community of math and biology educators who share resources and methods for preparing students to use quantitative approaches to tackle real, complex biological problems. The FMN group Make TRUBLE, or Make Teaching with *R* to Undergraduates Be Less

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her lesson, which focuses on an element of statistical analysis, can be found online .

Feburary 2020

Biology major **Rebecca Chastain** '20 presented a poster coauthored with Professor of Biology **Ben Pierce** at the Texas Academy of Sciences meeting in Nacogdoches, TX, February 28–29. Chastain's poster on Texas chirping frogs won first place in the terrestrial ecology section of the meeting.

January 2020

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research in genetics.

undergraduate research talks. The research on environmental DNA started during SCOPE 2018 and 2019 and will hopefully soon contribute to submissions to peer-reviewed journals. Their talk titles included the following:

- "Looking at and beyond the Horizon: Studying Nonnative Apple Snails in Texas and Diversity across the Globe" by Burks.
- "Indicating Invasion with eDNA: Detecting Apple Snails along Oyster Creek" by Muskara, coauthored by Shellsea Miller '20, Burks, and Matthew Barnes '06, assistant professor of natural resources management at Texas Tech University.
- "Old School or New School: Comparing the Efficiency of eDNA Sampling by Hand and with the ANDe™ eDNA Backpack" by Galassini, coauthored by Nyaberi, Burks, and Barnes.
- "Effect of the Degradation of eDNA in the Presence of Microplastics" by Nyaberi, coauthored by Galassini, Burks, and Barnes.

Professor of Biology **Ben Pierce**, the Williamson County Conservation Fund (WCCF), and Southwestern University cosponsored the symposium, which attracted 118 attendees and featured 25 talks.

Assistant Professor of Biology **Kim McArthur** was awarded the ADInstruments Educator Scholarship to attend the CrawFly neurophysiology course at the University of the Incarnate Word, January 9–12. This course provides intensive hands-on training for undergraduate educators developing laboratory courses in neuroscience to encourage integration of high-impact research experiences into the undergraduate curriculum. McArthur plans to develop a course in neurobiology that can incorporate modules from this training course.